

Day 116: Reinforcement Learning Fundamentals and Q-Learning Introduction

1. Day number

Day 116

2. Topic name

Reinforcement Learning and Very Basic Q-Learning

3. Connection

You have now seen three broad machine-learning ideas:

Supervised Learning
→ learn from labeled examples

Unsupervised Learning
→ discover patterns or structure

Reinforcement Learning
→ learn through actions and rewards

In reinforcement learning, we usually do not give the model a correct answer for every situation.

Instead, an **agent** interacts with an **environment**, takes actions, receives rewards, and gradually learns which actions tend to work well.

4. Important topics

Agent

The **agent** is the learner or decision-maker.

Examples:

Game character
Robot
Software player
Delivery bot

The agent chooses what action to take.

Environment

The **environment** is the world the agent interacts with.

For example, in a grid game:

```
+---+---+---+
|   |   | G |
+---+---+---+
|   |   |   |
+---+---+---+
| S |   |   |
+---+---+---+
```

S could be the starting position.

G could be the goal.

The grid is the environment.

State

A **state** describes the agent's current situation.

For a grid game, the state could simply be:

```
Current grid position
```

For example:

```
State A → bottom-left square
State B → bottom-middle square
State C → middle-middle square
```

Action

An **action** is something the agent can choose to do.

For a grid:

```
UP
DOWN
LEFT
RIGHT
```

The available actions can depend on the environment.

Reward

A **reward** is feedback from the environment after an action.

For example:

```
Reach goal      → +10
Hit dangerous area → -5
Normal move     → 0
```

Positive reward usually means:

That outcome was useful.

Negative reward usually means:

That outcome was undesirable.

Policy

A **policy** is the agent's strategy for choosing actions.

Conceptually:

```
If I am in State A → move RIGHT
If I am in State B → move UP
If I am in State C → move RIGHT
```

A policy answers:

What action should I choose in this state?

Episode

An **episode** is one complete attempt or run.

For example:

```
Start
↓
Move
↓
Move
↓
Move
↓
Reach goal
↓
Episode ends
```

Then the environment may reset and another episode begins.

Q-value idea

A **Q-value** represents how useful the agent currently thinks a particular action is in a particular state.

Think of it as:

```
Q(state, action)
```

For example:

```
Q(State A, RIGHT) = high
```

could mean:

Moving right from State A has worked well in past experience.

A low Q-value might mean the action has usually been less useful.

For Day 116, you do not need the full Q-learning equation.

5. Foundational notes

Reinforcement learning is different from ordinary supervised learning.

In supervised learning:

```
Input → correct label
```

For example:

```
Email → Spam  
Image → Cat  
House features → Price
```

The model learns from known answers.

In reinforcement learning:

```
State  
↓  
Agent chooses action  
↓  
Environment changes  
↓  
Reward received
```

The agent has to discover which actions lead to better long-term results.

This creates a new problem:

An action may not give an immediate reward but could still help the agent reach a good outcome later.

That idea becomes very important in reinforcement learning.

6. The reinforcement-learning interaction loop

The basic RL loop is:

```
Observe state
  ↓
Choose action
  ↓
Environment responds
  ↓
Receive reward
  ↓
Observe new state
  ↓
Learn
  ↓
Choose another action
```

Suppose a robot is in a maze.

Step 1: Observe the state

```
"I am in square A."
```

Step 2: Choose an action

```
Move RIGHT
```

Step 3: Environment changes

The robot moves into square B.

Step 4: Receive reward

Perhaps:

```
reward = 0
```

because nothing special happened.

Step 5: Continue

The robot makes another decision.

Eventually:

Move UP
→ reach goal
→ reward = +10

That positive reward gives the agent useful information about the actions that helped reach the goal.

7. Easy grid analogy

Consider this tiny environment:

```
+-----+-----+-----+  
|   A   |   B   |  GOAL  |  
+-----+-----+-----+
```

The agent starts at A.

Available action:

RIGHT

Possible journey:

```
A  
↓ RIGHT  
B  
↓ RIGHT  
GOAL
```

Suppose the rewards are:

```
A → B      reward = 0  
B → GOAL    reward = +10
```

At first, the agent may not know that moving right from B is useful.

After trying it:

```
State B  
Action RIGHT  
Reward +10
```

the agent learns:

RIGHT is a good action in State B.

So its estimate for:

Q(B, RIGHT)

can increase.

8. Q-table conceptually

A **Q-table** stores estimated Q-values for state-action combinations.

Suppose the environment has three states:

A
B
GOAL

and two possible actions:

LEFT
RIGHT

A simple Q-table might initially look like this:

State	LEFT	RIGHT
A	0	0
B	0	0
GOAL	0	0

At the beginning, the agent may know nothing.

So every value starts at:

0

Now suppose the agent reaches B and chooses:

RIGHT

and receives:

+10

The agent should now consider that action more useful.

Conceptually, the table might become:

State	LEFT	RIGHT
A	0	0

State	LEFT	RIGHT
B	0	5
GOAL	0	0

The exact value 5 is just an example.

The important idea is:

Good reward
↓
Increase confidence in that state-action choice

After more experience, values can continue changing.

9. Problem statement

Consider this tiny environment:

START → MIDDLE → GOAL

There are three states:

S0 = START
S1 = MIDDLE
S2 = GOAL

Possible actions:

LEFT
RIGHT

Suppose the rewards are:

Move normally → 0
Reach GOAL → +10
Move into danger → -5

Your task is to identify:

1. Who or what is the **agent**?
2. What is the **environment**?
3. What are the **states**?
4. What are the possible **actions**?
5. What rewards can the agent receive?
6. What does one **episode** look like?

7. If the agent is in S_1 , chooses RIGHT, reaches the goal, and receives +10, what should happen conceptually to $Q(S_1, \text{RIGHT})$?

Do not calculate the exact Q-learning update.

10. Concepts used

This exercise uses:

- reinforcement learning,
- agent,
- environment,
- state,
- action,
- reward,
- policy,
- episode,
- Q-value,
- Q-table,
- learning through experience.

A useful summary is:

```
Agent
  ↓
observes
  ↓
State
  ↓
chooses
  ↓
Action
  ↓
Environment changes
  ↓
Reward
  ↓
Agent learns
```

11. Thought process

When you see a beginner RL problem, think in this order.

Step 1: Identify the agent

Ask:

Who is making decisions?

Example:

Robot

Step 2: Identify the environment

Ask:

What world is the agent interacting with?

Example:

Grid or maze

Step 3: Identify the states

Ask:

What situations can the agent be in?

Example:

Position A
Position B
Position C

Step 4: Identify the actions

Ask:

What choices can the agent make?

Example:

LEFT
RIGHT
UP
DOWN

Step 5: Identify the rewards

Ask:

What feedback does the agent get?

For example:

```
Goal    → +10
Danger  → -5
Move    → 0
```

Step 6: Think about repeated experience

The agent may first make poor decisions:

```
LEFT
LEFT
Danger
Reward = -5
```

Later it may discover:

```
RIGHT
RIGHT
Goal
Reward = +10
```

Its estimates should gradually favor the more successful decisions.

Step 7: Think in state-action pairs

Instead of simply asking:

```
"Is RIGHT good?"
```

ask:

```
"Is RIGHT good in this particular state?"
```

That distinction is important.

An action can be useful in one state and poor in another.

12. Beginner-friendly pseudocode

```
START

create environment

create Q-table
```

```
set initial Q-values to 0

repeat for several episodes:

    place agent at starting state

    while episode is not finished:

        observe current state

        choose an action

        perform action

        receive reward

        observe next state

        update the Q-value conceptually:
            good result → value may increase
            bad result  → value may decrease

        move to next state

END
```

A slightly more detailed mental model is:

```
current state
  ↓
choose action
  ↓
receive reward
  ↓
see next state
  ↓
update Q(state, action)
  ↓
continue
```

The goal is that, after enough experience, the Q-table contains useful information about which actions tend to lead to better outcomes.

13. Easy edge cases

Edge case 1: Negative reward

Suppose:

State B
Action LEFT
→ dangerous square
→ reward = -10

The agent should learn that:

$Q(B, \text{LEFT})$

is less attractive.

Conceptually:

Negative reward
↓
Action becomes less desirable

This helps the agent avoid repeating poor decisions.

Edge case 2: No immediate reward

Suppose:

A → B
reward = 0

but then:

B → GOAL
reward = +10

Moving from A to B may still have been useful even though it produced no immediate reward.

Why?

Because reaching B made it possible to reach the goal afterward.

This is one of the important ideas in reinforcement learning:

Some useful actions matter because of future rewards, not immediate rewards.

You do not need the mathematics for this yet.

Edge case 3: Several actions seem equally good initially

If every Q-value starts at zero:

```
LEFT  = 0
RIGHT = 0
UP    = 0
DOWN  = 0
```

the agent initially has little information about which action is best.

It must gain experience.

That is why reinforcement learning usually involves some amount of **exploration**.

For Day 116, just understand exploration as:

Trying actions so the agent can learn what happens.

Edge case 4: Reward too easy to get

Imagine an environment where:

```
Standing still → +5
Reaching goal  → +10
```

The agent might repeatedly stand still because it already receives easy rewards.

This illustrates why reward design matters.

The reward should encourage the behavior we actually want.

14. Expected conceptual result

After repeated episodes, imagine the Q-table becomes:

State	LEFT	RIGHT
START	-1	4
MIDDLE	-2	8
GOAL	0	0

You do not need to calculate these values.

Conceptually, they suggest:

```
At START:
RIGHT looks better.

At MIDDLE:
RIGHT looks much better.
```

At GOAL:
episode is already finished.

The learned policy could therefore become:

START → RIGHT
MIDDLE → RIGHT
GOAL → stop

The complete behavior would be:

START
↓ RIGHT
MIDDLE
↓ RIGHT
GOAL

This is the basic idea behind Q-learning:

Learn useful values for state-action combinations from repeated rewards and use those values to make better decisions.

15. Hint only

For the problem, start by separating the pieces:

Who makes decisions?
→ agent

Where does it act?
→ environment

What situations exist?
→ states

What choices can it make?
→ actions

What feedback does it receive?
→ rewards

Then focus on this event:

State = MIDDLE
Action = RIGHT
Result = GOAL
Reward = +10

Ask:

After receiving a positive reward, should the agent consider RIGHT from MIDDLE more useful or less useful?

That tells you what should happen conceptually to:

$Q(\text{MIDDLE}, \text{RIGHT})$

For Day 116, keep this mental model:

```
State
↓
Choose action
↓
Receive reward
↓
Observe next state
↓
Update what was learned
↓
Repeat
```

You do **not** need deep reinforcement learning, neural networks, or the full Q-learning equation yet.